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A ADDITIONAL RELATED WORK

Mechanism Design for LLMs. This nascent area explores mechanisms for settings where multiple agents interact via LLMs. Duetting et al. [8] design token-level auctions where bids influence generated distributions, analyzing incentive compatibility. Dubey et al. [7] design auctions for incorporating ads into LLM summaries, using “prominence” as an intermediate allocation variable. Common ground exists in the high-level goal of aggregating inputs from multiple agents (represented algorithmically/via LLMs) to produce a collective textual output. However, our work differs significantly in methodology and objective. We do not employ economic mechanisms like auctions, bids, or payments. Instead, we formulate the aggregation problem as a multi-objective optimization within an MDP, aiming to achieve a fair consensus based on social choice criteria, rather than allocating influence or generating content based on bids.

B ILLUSTRATION OF DERIVING THE POLICY FROM THE LOTTERY

The relationship between the target distribution p^* over leaves and the calculation of the policy π^* at an internal node s is illustrated in Figure 5.

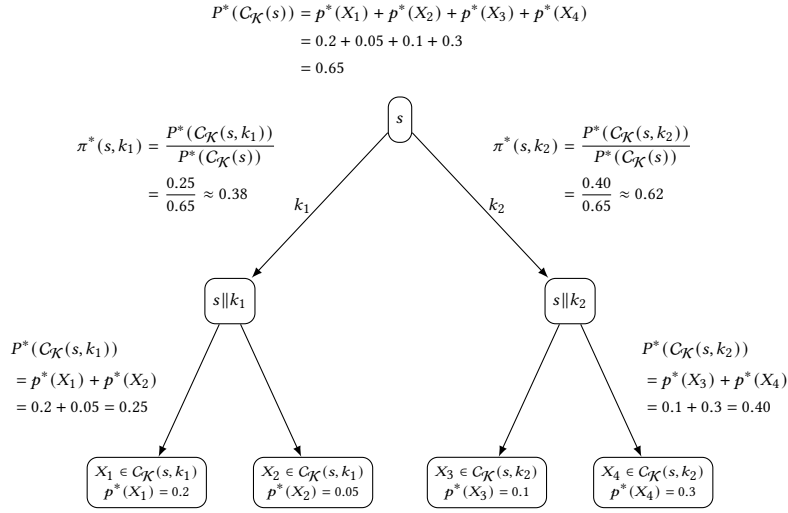


Figure 5: Illustration of the induced stochastic policy π^* at state s . The optimal lottery p^* assigns probabilities to the leaf nodes (complete paths). The probability $P^*(C_K(s))$ is the sum of $p^*(X)$ for all leaves reachable from s . The probability $P^*(C_K(s, k))$ is the sum for leaves reachable via action k . The policy $\pi^*(s, k)$ is the conditional probability of taking action k .

C DEFERRED PROOFS

C.1 Proof of Theorem 1

PROOF. We prove by induction on the depth of state s in the chunked tree that the probability of reaching state s under policy π^* , denoted $P_{\pi^*}(s)$, is equal to $P^*(C_K(s))$, the total mass assigned by p^* to leaves passing through s .

Base Case (Depth 0): The initial state is s_0 . $P_{\pi^*}(s_0) = 1$ by definition. Also, $C_K(s_0) = C_K$ (all paths pass through the start state), and $P^*(C_K(s_0)) = \sum_{X \in C_K} p^*(X) = 1$ since p^* is a probability distribution. Thus, $P_{\pi^*}(s_0) = P^*(C_K(s_0))$.

Inductive Hypothesis (IH): Assume that for all states s at depth d , $P_{\pi^*}(s) = P^*(C_K(s))$.

Inductive Step: Consider an arbitrary state s' at depth $d + 1$. State s' must be reached from a unique predecessor state s at depth d by taking a specific action (chunk) k , such that $s' = s||k$. The probability of reaching s' under π^* is:

$$P_{\pi^*}(s') = P_{\pi^*}(s) \cdot \pi^*(s, k)$$

$$= P^*(C_K(s)) \cdot \pi^*(s, k) \quad (\text{by IH})$$

If $P^*(C_K(s)) = 0$, then $P_{\pi^*}(s) = 0$, implying $P_{\pi^*}(s') = 0$. Also, if $P^*(C_K(s)) = 0$, then $P^*(C_K(s, k)) = 0$ since $C_K(s, k) \subseteq C_K(s)$. Since $s' = s||k$, $C_K(s') = C_K(s, k)$, so $P^*(C_K(s')) = 0$. Thus, $P_{\pi^*}(s') = P^*(C_K(s')) = 0$.

If $P^*(C_K(s)) > 0$, we use the definition of $\pi^*(s, k)$:

$$P_{\pi^*}(s') = P^*(C_K(s)) \cdot \frac{P^*(C_K(s, k))}{P^*(C_K(s))}$$

$$= P^*(C_K(s, k))$$

Since $s' = s||k$, we have $C_{\mathcal{K}}(s') = C_{\mathcal{K}}(s, k)$. Therefore,

$$P_{\pi^*}(s') = P^*(C_{\mathcal{K}}(s'))$$

This completes the inductive step.

Conclusion: The induction holds for all states s . Now, consider any leaf node $X \in C_{\mathcal{K}}$. A leaf node is a state at the maximum depth. The set of paths passing through leaf X is just the singleton set $\{X\}$, so $C_{\mathcal{K}}(X) = \{X\}$. Applying our proven result for state $s = X$:

$$P_{\pi^*}(X) = P^*(C_{\mathcal{K}}(X)) = P^*(\{X\}) = p^*(X)$$

Since this holds for all $X \in C_{\mathcal{K}}$, the distribution p_{π^*} induced by policy π^* is identical to the target distribution p^* . $\square$

C.2 Proof of Theorem 2

PROOF. Fix $R > 1$. Choose parameters $\varepsilon \in (0, 1)$ and $\delta \in (0, (1 - \varepsilon)^2/R)$. Consider two agents $i \in \{1, 2\}$ and two leaves x^* and x^- . Let $C = \{x^*, x^-\}$. Define utilities

$$u_i(x^*) = 1 - \varepsilon, \quad u_i(x^-) = \sqrt{\delta} \quad \text{for } i = 1, 2.$$

Assume chunking removes x^* and keeps x^- , so $K = \{x^-\}$.

First, optimize over C . For any lottery $p \in \Delta(C)$ let $\alpha = p(x^*) \in [0, 1]$. Each agent's expected utility equals

$$\sum_{x \in C} u_i(x)p(x) = \alpha(1 - \varepsilon) + (1 - \alpha)\sqrt{\delta} \leq 1 - \varepsilon,$$

since $\sqrt{\delta} < 1 - \varepsilon$. Hence

$$\text{NW}_C(p) = \left(\alpha(1 - \varepsilon) + (1 - \alpha)\sqrt{\delta} \right)^2 \leq (1 - \varepsilon)^2,$$

with equality only at $\alpha = 1$. Therefore $p_C^* = e_{x^*}$ and $\text{NW}_C(p_C^*) = (1 - \varepsilon)^2$.

Second, optimize over K . Any $p \in \Delta(K)$ satisfies

$$\sum_{x \in K} u_i(x)p(x) \leq \sqrt{\delta} \quad \text{for } i = 1, 2,$$

so $\text{NW}_K(p) \leq \delta$. With $K = \{x^-\}$ we have $p_K^* = e_{x^-}$ and $\text{NW}_K(p_K^*) = \delta$.

Combining the two bounds gives

$$\frac{\text{NW}_C(p_C^*)}{\text{NW}_K(p_K^*)} \geq \frac{(1 - \varepsilon)^2}{\delta} > R,$$

by the choice of δ . This establishes the claim. $\square$

D SEARCH ALGORITHMS

Algorithm 2 Finite Lookahead Egalitarian Welfare Maximization

Require: Set of agents N , Lookahead depth d , Branching factor B , Max length $L_{\max}$

```

1: Initialize current state  $s_0$  to the empty sequence;  $t \leftarrow 0$ 
2: Initialize generated path  $X_{fl} \leftarrow (s_0)$ 
3: while  $t < L_{\max}$  and  $s_t$  is not terminal do
4:   Let  $X_{prefix}$  be the path corresponding to  $s_t$ .
5:   Let  $\mathcal{P}_d(s_t)$  be the set of all paths  $P = (a_1, \dots, a_k)$  starting from  $s_t$  such that  $k \leq d$  and  $X_{prefix} \parallel P$  does not exceed length  $L_{\max}$ .
6:   Find a path  $P^* = (a_1^*, \dots, a_{k^*}^*) \in \mathcal{P}_d(s_t)$  that maximizes the lookahead objective:

$$\max_{P \in \mathcal{P}_d(s_t)} \min_{i \in N} U_i^{\log}(X_{prefix} \parallel P)$$

7:   if no path  $P^*$  found (e.g.,  $s_t$  is terminal) then
8:     Break
9:   end if
10:  Take the first action  $a^* \leftarrow a_1^*$ .
11:  Update state:  $s_{t+1} \leftarrow T(s_t, a^*)$ 
12:  Append  $a^*$  to the sequence represented by  $X_{fl}$ .
13:   $t \leftarrow t + 1$ 
14: end while
15: Perform brush up on  $X_{fl}$  (using prompt defined in Figure 8).
16: return Complete path  $X_{fl}$ 

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Algorithm 3 Egalitarian Welfare Beam Search

Require: Set of agents N , Beam width w , Branching factor B , Max length $L_{\max}$

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1: Initialize beam  $\mathcal{B}_0 = \{(s_0, \text{path } s_0)\}$  with the empty sequence path
2: for  $t = 0$  to  $L_{\max} - 1$  do
3:    $C_{t+1} \leftarrow \emptyset$  ▷ Candidate set for next beam
4:   for each path  $X_{path}$  represented by state sequence  $(s_0, \dots, s_t)$  in  $\mathcal{B}_t$  do
5:     if  $s_t$  is not terminal then
6:       Consider  $B$  possible next actions  $A_B(s_t)$  from state  $s_t$ 
7:       for each action  $a \in A_B(s_t)$  do
8:          $s_{t+1} \leftarrow T(s_t, a)$ 
9:          $X_{new\_path} \leftarrow X_{path} \parallel a$  ▷ Append action to path sequence
10:        Add  $X_{new\_path}$  (represented by its state sequence) to  $C_{t+1}$ 
11:      end for
12:    else
13:      Add  $X_{path}$  (already terminal) to  $C_{t+1}$ 
14:    end if
15:  end for
16:  For each path  $X \in C_{t+1}$ , compute its score  $f(X) = \min_{i \in N} U_i^{\log}(X)$ .
17:   $\mathcal{B}_{t+1} \leftarrow$  top  $w$  paths from  $C_{t+1}$  according to score  $f(X)$ .
18: end for
19: Select path from final beam  $\mathcal{B}_{L_{\max}}$  with the highest score  $f(X)$ .
20: Perform brush up on selected path (using prompt defined in Figure 8).
21: return Brushed up statement.

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E ADDITIONAL EMPIRICAL RESULTS

E.1 Credit Assignment

Table 2: Credit assignment results for Gemma 2 9b Instruction-Tuned. Darker green indicates larger Z-score. Z-score column is for altered tokens. Alterations are represented by "<misaligned>/<aligned>".

Reference policy prompt	User policy prompt	Sequence	Z-Score
User food profile: empty	User food profile: vegetarian	I am having <misaligned>chicken/tofu</misaligned> en chila das tonight . Then I am going to meet up with some friends .	3.69
User location profile: empty	User location profile: lives in a cold climate	I ’ m going to the beach/mountains this weekend to <misaligned>surf/ski</misaligned> . I need to buy some new clothes .	1.01, 2.86
User time profile: empty	User time profile: morning	I am about to eat some food . I am going to have <misaligned>spaghetti/pancakes</misaligned> . I will use my phone to order it .	3.13
User opinion: empty	User opinion: Favors stricter gun control laws.	Implementing background checks that are <misaligned>less/more</misaligned> strict for gun purchases is essential . Also , my favorite color is orange .	1.77

E.2 Supplemental Table for Figure 4

Table 3: Egalitarian Welfare (perplexity) across all scenarios shown in Figure 4.

Method	Scenario 1	Scenario 2	Scenario 3	Overall
Beam Search	3.22 ± 0.37	2.66 ± 0.15	2.74 ± 0.25	2.87 ± 0.37
Finite Lookahead	4.29 ± 0.66	3.83 ± 0.20	4.43 ± 0.67	4.18 ± 0.61
Best-of-N	8.57 ± 1.15	6.70 ± 0.32	3.77 ± 0.75	6.35 ± 2.14
Prompted Habermas Machine	10.58 ± 4.76	10.39 ± 2.14	8.03 ± 1.94	9.67 ± 3.42
Habermas Machine	17.18 ± 0.00	15.71 ± 0.00	14.18 ± 0.19	15.69 ± 1.23

E.3 Scenarios and Sample Consensus Statements from subsection 6.3

Table 4: Agent Opinions and Method Consensus on Scenario 1: "Are we living in a golden age of democracy?"

Source	Opinion / Consensus Statement
<i>Individual Agent Opinions</i>	
Agent 1	No, I think the golden age of democracy is long gone. I think a system where the first past the post wins is not working and we need to move to a model of proportional representation which would give more people the feeling that their voices were being heard. On the subject of voices, I'm strongly of the opinion that we have become a society where the loud voices of minority groups are able to impose their views on the rest of the population which to me is no democracy at all.
Agent 2	Worldwide democracy is more present than it's ever been in history. So yes, compared to previous ages in history I believe we are. Although that's not to say we can't improve - many countries still don't operate democratically, and in the ones that do, corruption is rife.
Agent 3	Yes, we are living in a golden age of democracy as democracy is of the people.
Agent 4	Not at all. The notion of democracy is being used for personal gains of those in government, and the system is manipulated. Around the world there is a considerable amount of oppression and lack of democratic values.
Agent 5	Compared to some parts of the world such as Russia and China which are actively regressing and reverting back to archaic ways of controlling their people, most Western countries are living through comparatively decent times, although problems still exist.
<i>Consensus Statements from Methods</i>	
Best of N	While acknowledging the imperfect nature of democracies worldwide, and recognising instances of regression and manipulation, we contend that, compared to historical periods and several non-democratic countries, the majority of Western countries and a sizeable portion of the global population currently enjoys a relatively high level of democratic freedoms; however, the overarching consensus leans towards a need for more inclusive and accountable forms of governance.
Finite Lookahead	Given the varying perspectives, we acknowledge that the definition of democracy can be subjective and dependent on geographical and cultural contexts, yet still, we recognize the existence of significant flaws in the current democratic systems worldwide, including instances of corruption, oppression, and manipulation, leading to the erosion of trust in governments and the democratic process. While some countries have made significant progress in consolidating democratic institutions, others have regressed, highlighting the need for continuous improvement and reform to ensure the preservation and strengthening of democratic values and principles. Therefore, we conclude that the current state of democracy worldwide is complex and multifaceted, with areas of improvement necessary to fully realize the ideals of democracy for all citizens, and acknowledge the presence of democratic systems in many parts of the world, particularly in the Western hemisphere, which, despite their imperfections, remain the best available option for promoting the common good and protecting individual rights and freedoms.
Prompted Habermas Machine	We acknowledge that the concept of a golden age of democracy is complex and contentious. While democracy has made significant progress in many parts of the world, we also recognize that fundamental challenges persist, including issues with representation, corruption, and the suppression of marginalized voices. These challenges highlight the need for a more critical examination of our democratic systems and a commitment to ongoing reform and improvement.
Habermas Machine	Although in the past we've had high hopes of a better world after the horrors of WWII and the fall of the Iron Curtain, democracy is in danger worldwide and may never reach its full potential. The Western world has poor democratic values, and even though democracy is spreading worldwide it is being overshadowed by the loud voices of minority groups.
Beam Search	We are not living in a golden age of democracy, but rather in an age where democracy is being challenged from within and from outside, and it is up to all of us to ensure that the principles of democracy are upheld.

Table 5: Agent Opinions and Method Consensus on Scenario 2: "Should the UK's ties to Europe be stronger or weaker?"

Source	Opinion / Consensus Statement
<i>Individual Agent Opinions</i>	
Agent 1	When we was in Europe we had good trade with them , The decision to leave was very bad for united kindom . We need to put the vote again to the British public i am sure this time the decision would be to remain
Agent 2	The natural evolution of our species has been to grow into ever bigger "tribes". Families ruled by their patriarchs became tribes ruled by elders became countries ruled by governments. It made sense that countries would evolve separately since they were geographically separate with no means of communication. Now our world is so connected, it is inevitable that we evolve into ever larger units such as the United States and the European Union. Eventually we will become a multi-planetary species ruled by an Earth government. To sever ties with Europe is a step in the wrong direction.
Agent 3	Although we did exit EU few years ago, we are geographically and economically in proximity to most EU countries. Several geographic, financial, political and economical parameters are intertwined with our neighbouring countries and, it would be advantageous to be in good relations to fully harness our economic, political, and financial facilities.
Agent 4	I believe the UK's ties to Europe should be stronger, as it would make trade deals easier to negotiate. This would allow us to benefit from a larger array of goods, which would make our imports cheaper. I also believe that closer ties with Europe in terms of immigration policies should occur as our immigration policy doesn't coincide with other nations around us. I believe a more united Europe would help all countries grow more through the movement of free labour, goods and services and more.
Agent 5	I think they should be stronger, as all the countries of Europe, except us, are part of the EU, and it makes Britain more isolated. Countries are stronger I believe when they work together with other countries and are open to the influence of other countries. I believe that leaving the EU was a step backwards and done for the wrong reasons. We are not the country we were 100 or even 50 years ago and the world is changing so much that we cannot afford to not be part of Europe.
<i>Consensus Statements from Methods</i>	
Best of N	Given the participants' opinions, we conclude that in order to achieve economic prosperity and streamline international relations, the UK's ties to Europe should be strengthened through collaborative efforts in trade, immigration, and economic policies while embracing the benefits of European integration, mutual understanding, and cooperation.
Finite Lookahead	The UK's ties to Europe should be strengthened to promote economic growth, facilitate trade, and foster greater unity among European nations. This can be achieved through mutually beneficial agreements and policies that reflect the shared interests of the UK and its European neighbours, leading to a more harmonious and prosperous region for all. Let's work together to build a stronger, more integrated Europe for the benefit of all its citizens, including the British people. Let's put the past behind us and look forward to a brighter, more collaborative future for the UK and Europe. Let's recognize the benefits of unity and work towards creating a more cohesive and successful Europe for generations to come.
Prompted Habermas Machine	We propose strengthening ties with Europe to foster greater economic cooperation, promote a more unified Europe, and address global challenges. However, we acknowledge that the path to this goal should include a reevaluation of the decision to leave the EU, allowing for a more informed and inclusive discussion about our relationship with Europe. Moreover, we recognize the long-term implications of European integration as part of a broader process of global unification, which may ultimately lead to a more interconnected and cooperative world.
Habermas Machine	The UK's ties to Europe should be stronger. This is because, although the UK did leave the EU, we are geographically and economically in proximity to most EU countries. Several geographic, financial, political and economical parameters are intertwined with our neighbouring countries and, it would be advantageous to be in good relations to fully harness our economic, political, and financial facilities.
Beam Search	The UK's ties to Europe should be stronger, and it would be beneficial for the country to be a part of the European Union or a similar entity that fosters cooperation and unity among its member states. This would allow for more efficient trade and economic growth, as well as greater collaboration on issues such as immigration, security, and environmental policy.

Table 6: Agent Opinions and Method Consensus on Scenario 3: "Is it okay to use animals for food?"

Source	Opinion / Consensus Statement
<i>Individual Agent Opinions</i>	
Agent 1	Yes, as human beings are natural meat eaters. However, it is morally wrong to keep animals in cruel conditions and use them for food. On the whole, farming is an unethical practise and many farming methods should be banned.
Agent 2	I think that it is ok to use animals for food because I enjoy eating meat. I don't think it is ok if the animal is endangered and I do think that there should be certain factors in place to ensure that they are well kept.
Agent 3	In my opinion it is ok to use animals for food. They provide nutrition and vitamins to use and I think it is just the basic food chain. I do understand why people choose to be vegetarian or vegan, but I do not agree with them trying to force their views upon others.
Agent 4	Humans have used animals for food since we first evolved as a species. Animals provide a great deal of nutrition that the human body needs to be healthy. It is certainly true that we now have more alternative foods available so that it can be possible to live healthily without consuming meat - though this does usually rely on taking nutritional supplements. I think that we should continue to diversify food production to provide people with meat alternatives, but also we should continue to improve the welfare of animals that are eaten for food and try to do this sustainably to minimise the impact this has on the animals wellbeing and on the land / environment too.
<i>Consensus Statements from Methods</i>	
Best of N	While considering the ethical and nutritional aspects of using animals for food, we agree that it is acceptable to consume meat in moderation, provided that animals are raised and slaughtered humanely, and that sustainable and environmentally-friendly farming methods are implemented to minimize harm.
Finite Lookahead	The use of animals for food is acceptable, however it should be done in a way that respects the welfare of the animals and considers the impact on the environment, with the aim of achieving sustainable food production methods that benefit both human and animal well-being, and also providing alternatives for those who choose not to consume meat. This includes improving farming practices, reducing the amount of waste and pollution, and promoting education on responsible food production and consumption practices. All sides of the issue should be taken into account, and the goal should be to achieve a balance between food security, animal welfare, and environmental sustainability.
Prompted Habermas Machine	We acknowledge that humans have traditionally used animals for food, and we recognize the importance of improving animal welfare and minimizing environmental impact to ensure sustainable practices. We understand that some consumers prefer eating meat, and we encourage diversifying food production options to include alternatives that can meet the nutritional needs of humans. Importantly, we emphasize the need to prohibit inhumane farming practices and actively work towards improving animal conditions, which aligns with our collective commitment to animal well-being and environmental stewardship.
Habermas Machine	Humans have used animals for food since we first evolved as a species. Animals provide a great deal of nutrition that the human body needs to be healthy. It is certainly true that we now have more alternative foods available so that it can be possible to live healthily without consuming meat - though this does usually rely on taking nutritional supplements. We should continue to diversify food production to provide people with meat alternatives, but also we should continue to improve the welfare of animals that are eaten for food and try to do this sustainably to minimise the impact this has on the animals wellbeing and on the land / environment too.
Beam Search	It is generally acceptable to use animals for food, as long as they are treated with respect and care, but we must also consider the impact of our food choices on the environment and animal welfare.

E.4 Additional Consensus Generation Experiments

We ran additional experiments on consensus generation. We restricted to scenarios with fewer than five agent opinions. We then applied k -means clustering ($k = 5$) to scenario embeddings from BAAI/bge-large-en-v1.5. Consensus statements were generated with Gemma 2 9B (instruction-tuned). We evaluated these statements using Egalitarian Perplexity (*EPPL*; lower is better), computed both in-model (the generating model, Gemma 2 9B Instruct) and out-of-model (Llama 3.1 8B Instruct). We also used an LLM-judge metric with GPT-4.1 (prompt in Figure 9). For each scenario, GPT-4.1 ranked candidates on behalf of each agent; we report the maximum (worst) rank across agents for each method (lower is better). All averages are over three random seeds.

E.4.1 Question 1: Scaling—Habermas vs. Best-of- N . We vary the number of candidates for Habermas and N for Best-of- N . Table 7 and Table 8 report means and standard deviations of *EPPL* (lower is better).

Table 7: Gemma *EPPL* (lower is better): Habermas vs. Best-of- N scaling.

Method	N	Mean	Std. Dev.
Habermas Machine	1	15.54	6.16
Habermas Machine	2	17.98	6.68
Habermas Machine	3	19.14	7.96
Habermas Machine	5	17.11	7.06
Habermas Machine	10	17.38	6.19
Habermas Machine	20	19.61	8.39
Habermas Machine	50	15.55	4.51
Best-of- N	1	18.59	11.35
Best-of- N	3	11.99	5.99
Best-of- N	5	9.75	5.29
Best-of- N	10	9.03	4.03
Best-of- N	20	6.92	2.27
Best-of- N	50	7.43	2.91

Table 8: Llama *EPPL* (lower is better): Habermas vs. Best-of- N scaling.

Method	N	Mean	Std. Dev.
Habermas Machine	1	12.20	7.39
Habermas Machine	2	12.05	2.87
Habermas Machine	3	13.69	2.38
Habermas Machine	5	12.48	4.80
Habermas Machine	10	11.73	2.56
Habermas Machine	20	13.37	2.29
Habermas Machine	50	12.82	3.81
Best-of- N	1	11.92	4.08
Best-of- N	3	9.14	3.33
Best-of- N	5	7.67	2.43
Best-of- N	10	7.67	2.43
Best-of- N	20	7.32	2.12
Best-of- N	50	7.58	2.58

Findings for RQ1. Across all N , Best-of- N yields lower *EPPL* than Habermas in both evaluations. Best-of- N improves as N increases, with marginal gains after $N = 20$. Habermas shows no clear improvement with more candidates. Under a fixed sample budget, Best-of- N reduces *EPPL* more effectively.

E.4.2 Question 2: Beam Search Scaling. We examine how beam width affects performance.

Table 9: Beam width scaling (average over 5 scenarios, 3 seeds each). Lower is better for all metrics.

Beam Width	Gemma <i>EPPL</i>		Llama <i>EPPL</i>		LLM Judge Max Rank	
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
2	10.01	1.95	10.07	1.70	3.13	0.80
4	7.84	1.33	10.16	1.99	2.80	0.61
6	10.54	10.40	9.20	2.51	3.13	0.38
8	12.56	7.47	12.69	5.90	3.93	0.15

Findings for RQ2. The optimal width depends on the metric. Width 4 is best on average for Gemma *EPPL* and the LLM-judge rank; width 6 is best for Llama *EPPL*. Width 8 degrades performance, likely due to myopic pruning based on partial *EPPL* scores within the beam search.

E.4.3 Question 3: Method Comparison. We compare Finite Lookahead ($d = 3$, $B = 3$), Beam Search (width = 4), Best-of- N ($N \in \{5, 10, 50\}$), and Habermas (candidates = 5).

Table 10: Overall comparison: LLM-judge max rank (lower is better).

Method	Max Rank	Std. Dev.	Min Max Rank	Max Max Rank
Best-of- N ($N = 50$)	3.33	0.72	2.00	4.00
Best-of- N ($N = 10$)	4.33	0.72	3.00	5.00
Beam Search (width = 4)	5.07	0.80	4.00	6.00
Best-of- N ($N = 5$)	5.33	0.72	4.00	6.00
Habermas (candidates = 5)	5.40	0.74	4.00	6.00
Finite Lookahead (depth = 3)	5.80	0.41	5.00	6.00

Table 11: Overall comparison: Gemma *EPPL* (lower is better).

Method	Mean	Std. Dev.	Min	Max
Best-of- N ($N = 50$)	7.43	2.91	3.92	10.96
Beam Search (width = 4)	7.84	1.33	6.34	9.89
Finite Lookahead (depth = 3)	8.01	2.23	5.60	10.97
Best-of- N ($N = 10$)	9.03	4.03	3.33	14.52
Best-of- N ($N = 5$)	9.75	5.29	4.72	18.42
Habermas (candidates = 5)	17.11	7.06	9.44	32.49

Table 12: Overall comparison: Llama *EPPL* (lower is better).

Method	Mean	Std. Dev.	Min	Max
Best-of- N ($N = 50$)	7.58	2.58	3.85	10.79
Best-of- N ($N = 5$)	7.67	2.43	4.76	10.95
Best-of- N ($N = 10$)	7.67	2.43	3.69	10.17
Finite Lookahead (depth = 3)	8.49	3.29	5.56	13.89
Beam Search (width = 4)	10.16	1.99	7.96	13.06
Habermas (candidates = 5)	12.48	4.80	6.48	23.30

Findings for RQ3. Best-of- N with large N ($N = 50$) is strongest on both *EPPL* metrics and on worst-case LLM-judge rank. Beam Search with width 4 is competitive on Gemma *EPPL* (second to Best-of- N with $N = 50$) and outperforms Best-of- N with $N \in \{5, 10\}$ on that metric. Finite Lookahead performs moderately well. Habermas (candidates = 5) lags on both *EPPL* and the judge metric. Method choice trades off average *EPPL*, worst-case judge rank, and compute budget. Token-level MDP methods (Beam Search and Finite Lookahead) also admit analysis within the MDP framework, which is useful for theory.

E.4.4 Bowling Green/Warren County (41 agents; Llama 3.1 8B Instruct). We evaluated the Bowling Green/Warren County data from Boehmer et al. [3]. Using 41 agents and one seed with Llama 3.1 8B Instruct, Table 13 reports Egalitarian Perplexity (*EPPL* calculated using Llama 3.1 8B Instruct; lower is better) and the full consensus statement for each method.

Table 13: Bowling Green/Warren County results (41 agents, one seed). Model: Llama 3.1 8B Instruct. Lower *EPPL* is better.

Method	<i>EPPL</i> ↓	Full consensus statement
Beam Search (width = 4)	3.29	To make Bowling Green/Warren County a better place to live, work, and spend time, we need to address our community’s most pressing issues, including a lack of accessible and affordable housing options, inadequate public transportation, and insufficient mental-health resources. Furthermore, we need to focus on improving our community’s infrastructure, such as upgrading our roads and traffic-management systems. Additionally, we must ensure our community is inclusive and welcoming to all residents, regardless of their background, culture, or socioeconomic status.
Best-of-4	6.17	To make Bowling Green/Warren County a better place to live, work, and spend time, we recommend prioritizing initiatives that expand high-speed internet access, improve traffic flow and safety, enhance public transportation, and enrich community life, while also prioritizing education, public health, and economic equality.
Prompted Habermas Machine (4 candidates + 1 critique)	14.08	To make Bowling Green/Warren County a better place to live, work, and spend time, we need to address pressing issues such as increasing high-speed internet access, improving traffic flow and safety by installing traffic cameras, increasing arts-education opportunities, and protecting tenant and renters’ rights. We also need to ensure welcoming and inclusive initiatives to support refugees and immigrants in broader community development. Invest in public transportation and enforce safety regulations such as parking policies to foster a sustainable environment. By working together to implement these actionable plans and initiatives, we can build a more comprehensive strategy to address the community’s high-priority needs.

F PROMPTS

```
You are generating a consensus statement that represents the views of multiple participants.

Your task is to continue the statement in a way that addresses the issue and considers all participants' opinions. Be concise and keep the statement short (less than 50 tokens) and focused. ONLY WRITE THE STATEMENT AND NOTHING ELSE.

Issue:
<issue>

Participants' opinions:
<opinion_1>
.
.
.
<opinion_n>

Consensus statement:
```

Figure 6: Reference policy prompt.

```
You are generating a statement that represents the views of a single participant.

Your task is to continue the statement in a way that addresses the issue and considers ONLY this participant's opinion. Be concise and keep the statement short (less than 50 tokens) and focused. ONLY WRITE THE STATEMENT AND NOTHING ELSE.

Issue:
<Issue>

Participant's opinion:
<opinion>

Statement reflecting ONLY this participant's opinion:
```

Figure 7: Agent policy prompt.

```
You are helping to fix ONLY the ending of a generated statement.

VERY IMPORTANT: If the statement ending is already complete and well-formed, DO NOT modify it at all.

Your task is to:
1. DO NOT change any part of the statement except the last few sentences if they have issues
2. Look for and fix ONLY these issues at the end of the statement:
  - Remove repetition in the final sentences
  - Complete any unfinished final sentence that can be completed easily
  - Remove any incomplete final sentence that cannot be meaningfully finished
3. Keep the changes minimal and focused only on the ending
4. DO NOT add any new information or opinions
5. DO NOT modify anything except problematic sentences at the end
6. If the statement is already complete and well-formed, return it EXACTLY as provided

Here is the statement:

<statement>
```

Figure 8: Brush up prompt.

You are evaluating consensus statements from the perspective of a specific agent. Your task is to rank multiple statements based on how well they represent the agent's opinion and interests on a given issue. Use ONLY the agent's stated opinion to determine the ranking.

Issue:
<issue>

Agent <agent_id>'s Opinion:
<agent_opinion>

Statements to Rank
Statement 1:
<statement_1_text>

Statement 2:
<statement_2_text>

...
Statement <n_statements>:
<statement_n_text>

Task:
From Agent <agent_id>'s perspective, rank all statements from most favorable (1) to least favorable (<n_statements>) based on how well they represent the agent's opinion and interests.

Provide your ranking as a JSON object with:
1. 'reasoning': brief explanation for your ranking decisions
2. 'ranking': an array of statement numbers in ranked order (best to worst)

For example: {'reasoning': 'Statement 3 best represents the agent's concerns about...',
'ranking': [3, 1, 2]}

Figure 9: Prompt used for LLM judge in subsection E.4.